IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

HVDC TECHNOLOGY AND CONTROL

Wednesday 17 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

Question 1 and 2 carries 25 marks each, Question 3 carries 30 marks

Question 1

a) Use of a voltage-dependent current order limit (VDCOL) within the control system of an LCC-HVDC link is prompted by the fact that a reduction in AC-side voltage at one end of the LCC-HVDC link leads to an increase of reactive power consumption at the opposite end. Explain why this happens by considering AC-side voltage reduction at the rectifier and inverter ends separately. Assume the usual control mode under normal operation at the two ends. You can explain qualitatively without using any expression.

[8]

b) Two separate 400 kV AC systems (System #1 and System #2) are to be interconnected using a bipole LCC-HVDC link. The short circuit levels at the intended point (bus) of connection within System #1 and System#2 are 35 GVA and 37 GVA, respectively. Calculate the maximum possible rated power capacity (in GW) of the LCC-HVDC link if the reactive power consumption by the converter station at each end is 55% of the rated active power flow through the DC link and the minimum stipulated effective short circuit ratio is 4. The equivalent impedance of both AC systems can be assumed to be purely reactive.

[4]

c) In the future, there is a plan to first build a 400 kV AC transmission line and then a second bipole LCC-HVDC link between System #1 and System #2 in part 1(b). The equivalent reactance of the AC transmission line is expected to be somewhere between 6 Ω and 7 Ω depending on the actual route. Calculate the maximum possible rated power capacity (in GW) of the second LCC-HVDC link if the reactive power consumption by the converter station at each end is 55% of the rated active power flow through the DC link and the minimum stipulated effective short circuit ratio is 4 as in part 1(b).

[8]

d) Considering the rated power capacities of the two LCC-HVDC links obtained in parts 1(b) and 1(c), determine the maximum allowable reactive power (both in GVAr and as a percentage of the rated active power transfer through HVDC) consumption by the LCC-HVDC converter station(s) connected to System #2 without violating the stipulated ESCR i) without and ii) with the AC transmission line in operation. Why percentage values are higher than the 55% figure specified in parts 1(b) and 1(c)?

[5]

Question 2

Two separate 400 kV AC systems are interconnected by a 2.2 GW, ±600 kV bipole LCC-HVDC cable link using the standard 12-pulse converter arrangement in each pole. The converter parameters are given in Table 2.1.

Nominal gate delay angle	16°
Minimum gate delay angle	3°
Minimum extinction advance angle	18°
Resistance of each pole	5 Ω
Commutation resistance for each 6-pulse converter	5 Ω
Current margin	18%

Table 2.1: Parameters of the converter at each end and the HVDC link

Under normal condition, the rectifier end of the HVDC link maintains the rated direct current while the inverter end maintains the rated direct voltage. Using calculations on a per-pole basis determine (a) the effective turns ratio of the converter transformers at the rectifier and inverter ends and (b) the reactive power (in MVAr) consumed at the rectifier and inverter ends for the following three scenarios:

i) Normal condition with rated AC system voltage at both ends.

[10]

ii) Inverter end AC system voltage at 90% of its rated value with rated AC system voltage at the rectifier end.

[5]

iii) Rectifier end AC system voltage at 85% of its rated value with rated AC system voltage at the inverter end. If the minimum gate delay angle is reached, assume that the rectifier continues to operate at the minimum gate delay angle.

[10]

Question 3

a) Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.1. The parameters of the converter are given in Table 3.1. Answer the questions below.

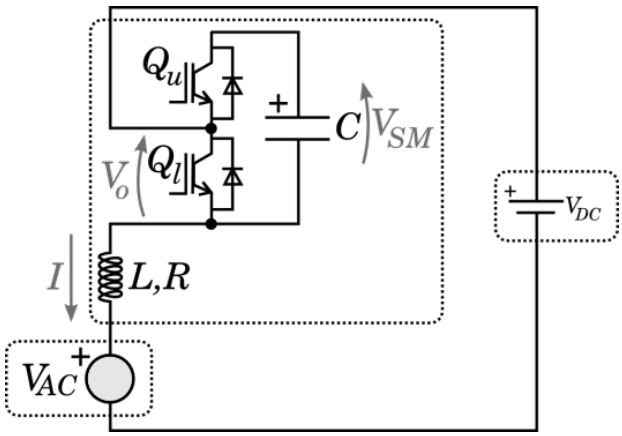


Figure 3.1 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	500	V	Rated DC grid voltage.
V_{AC}	230	V rms	Rated AC grid voltage
f_{rated}	50	Hz	Rated frequency of the AC grid
R	0.3	Ω	Equivalent resistance of bridge plus the filter
X	2.5	Ω	Inductance of the filter

Table 3.1 – Parameters of the one-arm modular AC/DC converter

i) What are the AC and DC components of the output voltage of the arm, V_o, when the output current, I, is zero? What is the minimum submodule capacitor voltage, V_{SM}, required to generate this voltage?

[3]

ii) Consider a stable operating point with zero net power balance across the submodule (i.e. its energy doesn't drift over time) and 3 kW of active power being delivered to the AC grid with unity power factor (P_{AC}=3 kW, Q_{AC}=0). Answer the following questions and highlight any assumptions or simplifications you make:

a. Explain what condition must be met for the submodule to have zero net power balance.

[3]

b. What are the DC and the AC components of the current required?

[4]

c. What are the AC and DC components of the output voltage of the arm, V_o, under this operating point?

[4]

iii) The diagram shown in Figure 3.2 is used to analyse the design of a current controller for this converter. Answer the questions below.

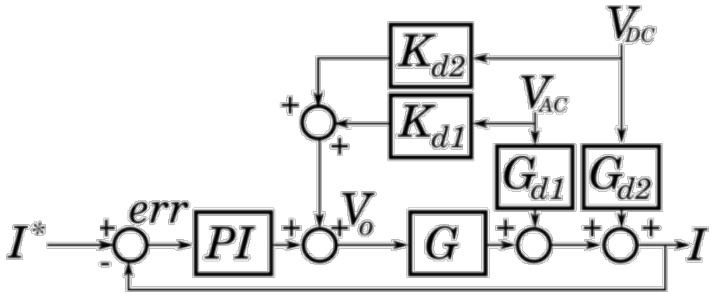


Figure 3.2 – Signal flow diagram for the analysis of the current controller

a. Write down the Laplace transfer functions for G_{d1} and G_{d2}.

[3]

b. Is a PI a good choice for the closed-loop current control of this converter?

[2]

b) Answer the following question related to the AC-DC converter shown in Figure 3.3. The converter parameters are summarised in Table 3.2.

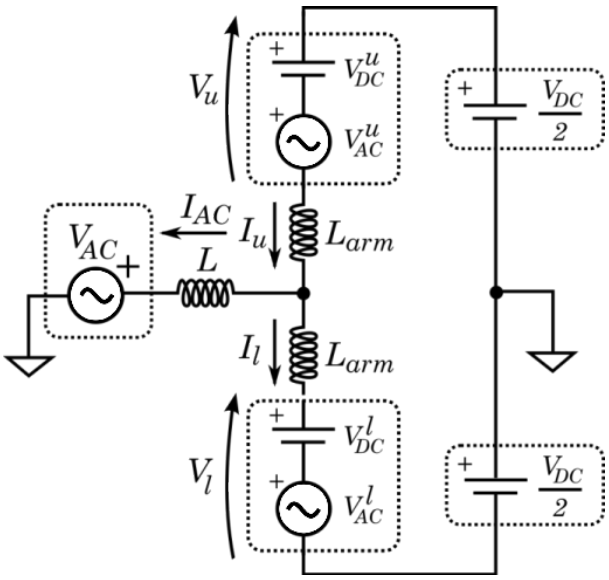


Figure 3.3 – Equivalent circuit of a two-arm single-phase AC/DC modular multilevel converter

Parameter	Value	Description
V_{DC}	120 kV	DC network voltage.
V_{AC}	38 kV	AC network voltage (RMS).
f_{AC}	50 Hz	Frequency of the AC network
S_{rated}	200 MVA	Rated power.
X_{arm}	1.4 Ω	Arm reactor inductance.
X_L	0.36 Ω	Phase reactor inductance
R_{arm}, R_L	0.07 Ω	Equivalent series resistances

Table 3.2 – Parameters of the two-arm single-phase AC/DC modular multilevel converter.

i) Calculate the impedances of the converter (ie arm and phase reactor, equivalent series resistances) in pu. Explain if the values look reasonable and what would happen if they were much larger.

[4]

ii) We are told that the currents of the arms of the converter are: $\bar{I}_{AC}^{u}$=25 A, I_{DC}^{u}=15 A, $\bar{I}_{AC}^{l}$= -10 A, $I_{DC}^{l} = 35\ A$. Calculate the AC and DC components of the summative current and the current going to the AC grid. Briefly explain the role of the summative current in the operation of the converter.

[3]

iii) We are told that at a specific point in time, the active power delivered by the converter to the AC grid is of 100 MW, the power drawn from the DC grid is 105 MW, the active power delivered by the upper stack and the lower stack of submodules using AC current is 60 MW and 40 MW respectively. Assume that there's no DC pole voltage imbalance in the DC grid and no DC current being fed to the AC grid. Assume that the losses due to the resistance of the converter are negligible and answer the following questions:

a. How much power do the upper stack and the lower stack exchange using DC current?

[2]

b. Does this correspond to a stable operating point? Explain the active power balance of the converter in this operating point.

[2]